

The core of our entire system is the products table. This will hold all the static information about a product. We also need to handle product variations (like size or color), categories, and brands.

Schema Design

Here are the table definitions for the Inventory Management module.

Table 1: categories

This table stores product categories to help with organization and reporting.

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the category.
name	VARCHAR(255)	NOT NULL, UNIQUE	Name of the category (e.g., "T-Shirts", "Electronics").
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of creation.
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of the last update.

Table 2: brands

This table stores brand information.

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the brand.
name	VARCHAR(255)	NOT NULL, UNIQUE	Name of the brand (e.g., "Nike", "Samsung").
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of creation.
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of the last update.

Table 3: products

This is the main "parent" product table. If a product has no variations, its details will be here. If it has variations, this acts as the template.

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the parent product.
name	VARCHAR(255)	NOT NULL	The main name of the product (e.g., "Polo T-Shirt").
description	TEXT	NULL	Detailed description of the product.
category_id	BIGINT	FOREIGN KEY (categories.id)	Links to the categories table.
brand_id	BIGINT	FOREIGN KEY (brands.id)	Links to the brands table.
hsn_code	VARCHAR(50)	NULL	HSN code for tax purposes.
tax_rate	DECIMAL(5, 2)	NOT NULL DEFAULT 0.00	Tax rate in percentage (e.g., 18.00).
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	Status to enable or disable the product.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of creation.
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of the last update.

Table 4: product_variants

This table holds the specific, sellable variations of a product. This is where inventory is actually tracked.

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the variant.
sku	VARCHAR(100)	NOT NULL, UNIQUE	Stock Keeping Unit. The most critical field.

product_id	BIGINT	FOREIGN KEY (products.id)	Links this variant to its parent product.
attributes	JSON	NULL	Stores variations (e.g., {"size": "M", "color": "Blue"}).
procurement_cost	DECIMAL(10, 2)	NOT NULL	Cost of Goods Sold for this specific variant.
supplier_lead_time	INT	NULL	Average time in days to get new stock.
warehouse_location	VARCHAR(100)	NULL	Physical location in the warehouse (e.g., "A-05-02").
stock_on_hand	INT	NOT NULL DEFAULT 0	The total physical quantity available.
stock_committed	INT	NOT NULL DEFAULT 0	Quantity allocated to unpaid/unshipped orders.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of creation.
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of the last update.

Table 5: inventory_logs

This is an append-only table for auditing every single change to inventory. This is crucial for tracking and accountability.

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique log entry ID.
variant_id	BIGINT	FOREIGN KEY (product_variants.id)	Which variant's stock was changed.

quantity_changed	INT	NOT NULL	The change amount (e.g., -1 for a sale, +50 for new stock).
new_stock_on_hand	INT	NOT NULL	The stock level <i>after</i> the change was applied.
change_type	VARCHAR(50)	NOT NULL	e.g., "SALE", "RETURN", "MANUAL_ADJUSTMENT".
reason	VARCHAR(255))	NULL	User-provided reason for manual adjustments.
order_id	BIGINT	NULL	If the change was due to a sale/return, link to the order.
user_id	BIGINT	NULL	ID of the user who made a manual change.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of the event.